



Forensic identification of *Agkistrodon halys* bite by the combination of ELISA and immunohistochemistry: a case report and literature review

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Abstract

Snakebites constitute a significant global public health concern. Based on various toxic components and action mechanisms, snake venoms are typically classified into hemotoxic, neurotoxic, and histotoxic types. The pathological changes caused by different toxicities vary, which is often one of the key points in the forensic identification of snake bites. However, relying solely on this aspect often has certain limitations in the identification of snake bite cases, with poor qualitative and quantitative results. This article presents a case of identifying death caused by an *Agkistrodon halys* bite through the combined application of ELISA and immunohistochemistry (IHC) methods. The concentration of snake venom in the subcutaneous tissue at the fang marks was precisely measured at 6.58 ng/mL. In addition, we summarized the forensic characteristics of such snake venom poisoning by a retrospective analysis of the literature, aiming to provide reference for the forensic identification.

Highlights

- A rare death caused by an *Agkistrodon halys* bite was identified and presented;
- A method for snake venom detection by a combination of ELISA and IHC was proposed.
- The forensic characteristics of *Agkistrodon halys* venom poisoning were summarized.

Keywords Forensic pathology · Snake bite · *Agkistrodon halys* · ELISA · Immunohistochemistry

Introduction

Snakebites represent a significant global health issue, with an estimated annual incidence of 5.4 million cases and 138,000 deaths worldwide [1]. Depending on the toxicity, snake venoms are usually divided into hemotoxic,

neurotoxic, and histotoxic types [2] causing various pathological changes [3, 4]. *Agkistrodon halys* is a common venomous snake in China. Its venom is usually hemotoxic, and its main toxic components are phospholipase A₂ (PLA₂), snake venom metalloproteinase (SVMP), and snake venom serine proteinase (SVSP) [5]. These components mainly

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cause pathological changes characterized by hemorrhage, edema, and necrosis, whose damage mechanisms included vascular basement membrane damage, coagulation factor depletion, and platelet aggregation inhibition [3, 6].

As for the forensic identification of snakebites, the case history, fang marks, and characteristic clinical manifestations with gross pathological changes were significant. However, the role that these information can play in the identification of snake venom types and the determination of their contents is limited. In our paper, the combination of ELISA and immunohistochemistry methods was applied to directly conduct qualitative and quantitative analysis of the snake venom, which helped confirming the snake as *Agkistrodon halys*. Also, we reviewed the forensic characteristics of such snake venom poisoning, aiming to provide assistance for the forensic identification of similar cases.

Case report

Case history

A 60-years-old male farmer was suspected of being bitten by a venomous snake on the inner arch of the right foot when working in the fields. However, due to poverty and a lack of knowledge, he did not receive hospital treatment for the next few days, instead, he received bloodletting and simple bandaging from a non-professional medical practitioner with limited medical knowledge. Unfortunately, the man died four days after the bite. An autopsy was performed six days after the death.

Autopsy findings

Two suspected fang marks were found on the inner side of the right foot arch, separated by an approximately distance of 0.3 cm, and there were multiple bloodletting incisions scattered around the suspected bite marks (Fig. 1a). There was extensive swelling from the inner side of the right

foot arch to the inner side of the right knee, with a dark red discoloration (Fig. 1b). The subcutaneous tissue exhibited extensive hemorrhage, edema, and necrosis.

The liver weighed 1320 g, and its cut surface showed congestion. The two kidneys weighed 310 g, with a dark red surface. The boundary between the cortex and medulla of the cut surface was indistinct, accompanied by pronounced congestion. The lungs presented with congestion and edema, with focal and patchy hemorrhagic areas on the cut surface. The pancreas displayed multiple patchy hemorrhages on both its surface and cut surface. No abnormalities were observed in other organs.

Histopathological findings

The subcutaneous tissue of the suspected fang mark area on the right foot arch had a large-area infiltration of inflammatory cells, with nuclear pyknosis, fragmentation, and dissolution, showing acute cellulitis (Fig. 2a). Some inflammatory cells were evenly distributed around the blood vessels, suggesting perivascularitis (Fig. 2b). A majority of the skeletal muscle fibers displayed cell necrosis and nuclear dissolution (Fig. 2c). The subcutaneous stroma exhibited edema accompanied by extensive hemorrhage and substantial fibrin exudation. Endothelial cells of some small blood vessels were swollen, with vessel walls exhibiting fibrinoid necrosis, and some small vessels contained hyaline thrombi and purulent formed.

The central region of the liver lobules exhibited hemorrhage, hepatocyte necrosis, dilation and congestion of the liver sinusoids, and interstitial hemorrhage in certain portal areas accompanied by inflammatory cell infiltration, indicative of toxic hepatitis (Fig. 2d).

In the kidneys, the epithelial cells of the proximal convoluted tubules were edematous and necrotic, with renal tubules containing casts within their lumens. The interstitial tissue displayed fibrosis with focal aggregation of monocytes and lymphocytes, and thrombi were observed in small blood vessels, indicating toxic nephropathy (Fig. 2e).

Fig. 1 The inner side of the right foot arch **a** Two suspected fang marks were found on the inner side of the right foot arch, separated by an approximately distance of 0.3 cm. **b** The subcutaneous tissue exhibited extensive hemorrhage, edema, and necrosis from the inner side of the right foot arch to the inner side of the right knee, with a dark red discoloration

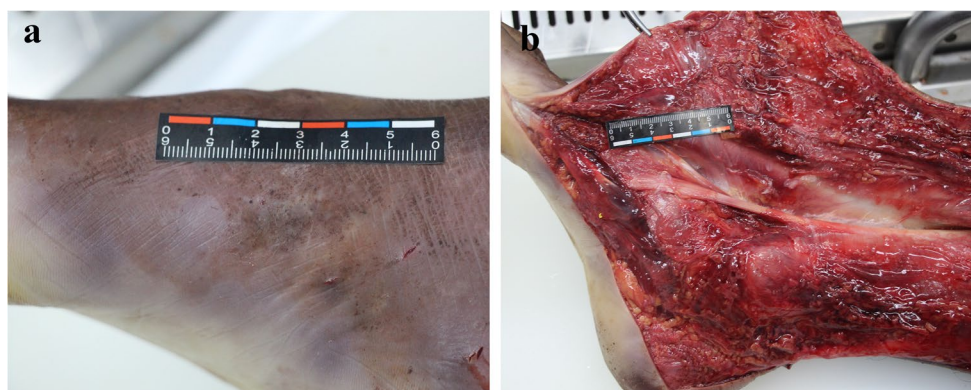
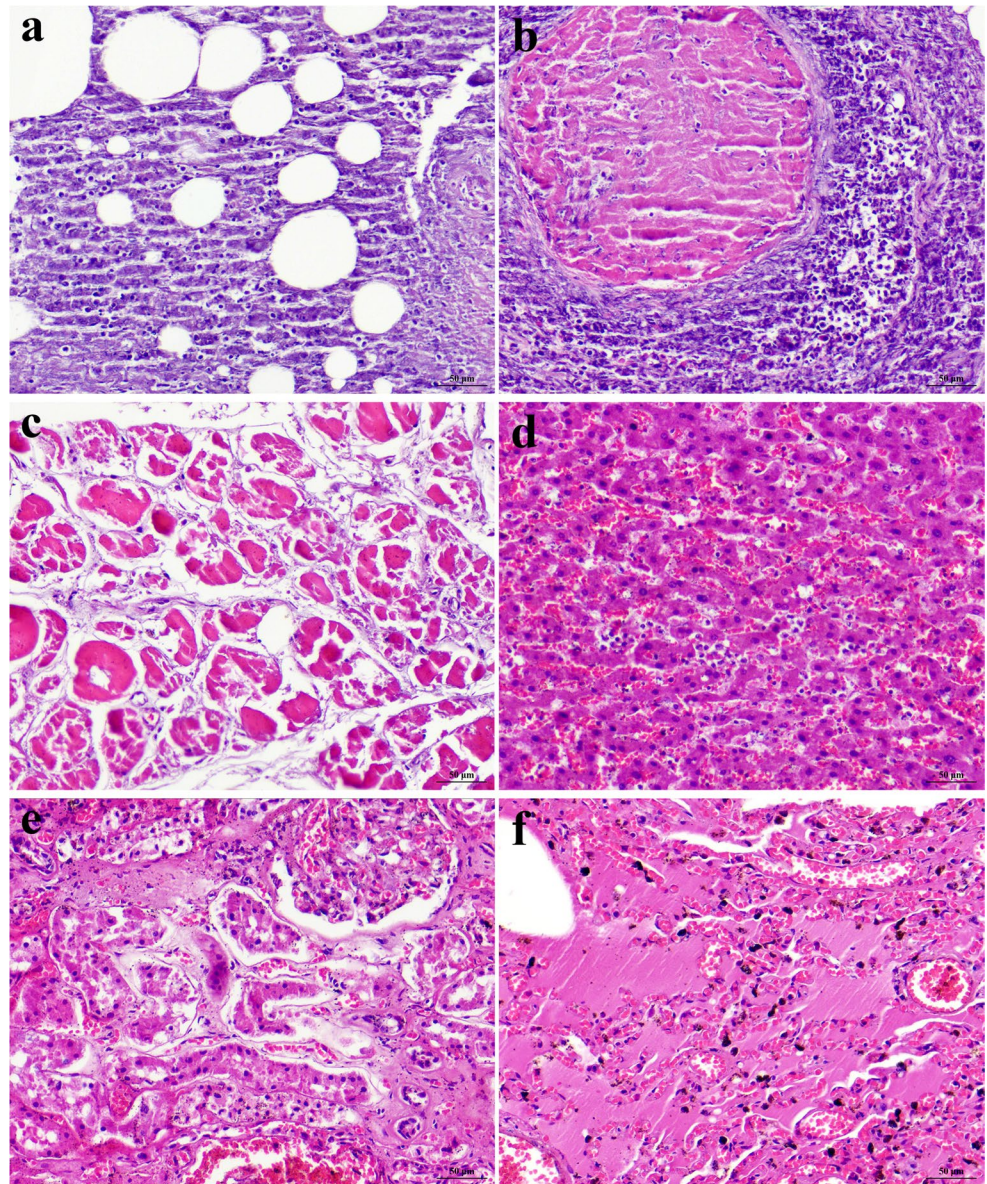


Fig. 2 Histopathological findings of the corpse **a** a large - area inflammatory cells infiltration mainly composed of neutrophils in the subcutaneous tissue, H&E staining, bar = 50 μ m. **b** some inflammatory cells were evenly distributed around the blood vessels, H&E staining, bar = 50 μ m. **c** the skeletal muscle fibers under the fang mark displayed nuclear dissolution and cell necrosis, H&E staining, bar = 50 μ m. **d** hemorrhage, hepatocyte necrosis, dilation and congestion of the liver sinusoids, and interstitial hemorrhage in certain portal areas of liver, H&E staining, bar = 50 μ m. **e** the epithelial cells of the proximal convoluted tubules were edematous and necrotic, H&E staining, bar = 50 μ m. **f** pulmonary edema and pulmonary interstitial small vessel congestion, H&E staining, bar = 50 μ m



The lungs, pancreas, and adrenal glands demonstrated marked hemorrhage and congestion (Fig. 2f), while no significant pathological alterations were observed in other organs.

Laboratory tests

Sandwich-ELISA Test For common venomous snake species in China, such as *Deinagkistrodon acutus* (*D. autus*), *Agkistrodon halys*, *Bungarus multicinctus* (*B. multicinctus*), and *Naja atra*, the previously established ELISA detection method for snake venom was refined [7]. In this case, we analyzed the type and concentration of snake venom in some tissue and organs.

Results *Agkistrodon halys* venom was detectable and quantified in all the samples tested: liver 12.56 ng/mL, lung 12.56 ng/mL, kidney 15.19 ng/mL, skin 16.57 ng/mL, muscle 11.49 ng/mL, subcutaneous tissue 6.58 ng/mL. *D. autus*, *B. multicinctus*, and *Naja atra* venom was not detected (Fig. 3).

Immunohistochemistry HRP-labeled horse-anti-*Agkistrodon halys* polyclonal antibody and SABC method was adopted for the detection of snake venoms. Results showed a positive reaction in the subcutaneous sweat gland duct cells of the suspected bite mark (Fig. 4a and b).

Other Toxicological Tests Routine toxicological tests were conducted, including for alcohol, rodenticides, organophosphate pesticides, barbiturates, benzodiazepines, and

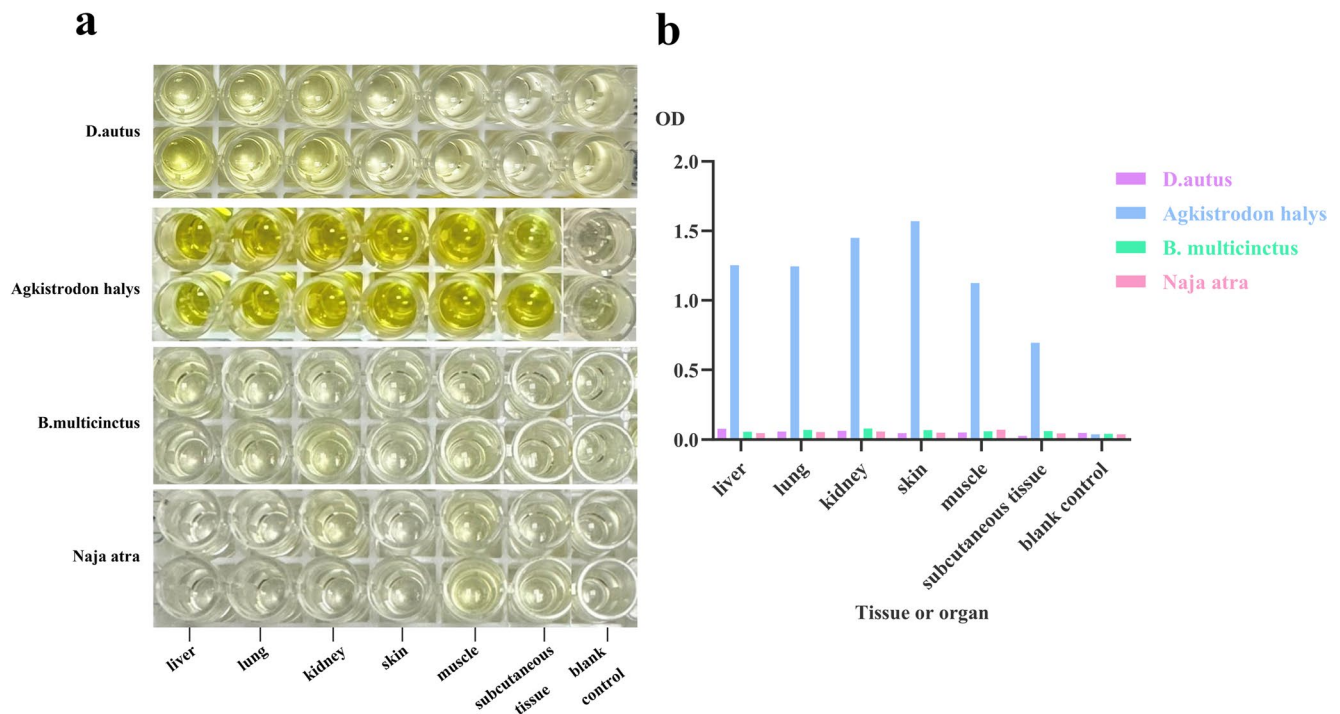


Fig. 3 Sandwich-ELISA test of the tissue and organs **a** The color development is shown for each tissue or organ for 4 venoms. **b** graphs for optical densities (at λ 450 nm) are shown for each tissue or organ for 4 venoms

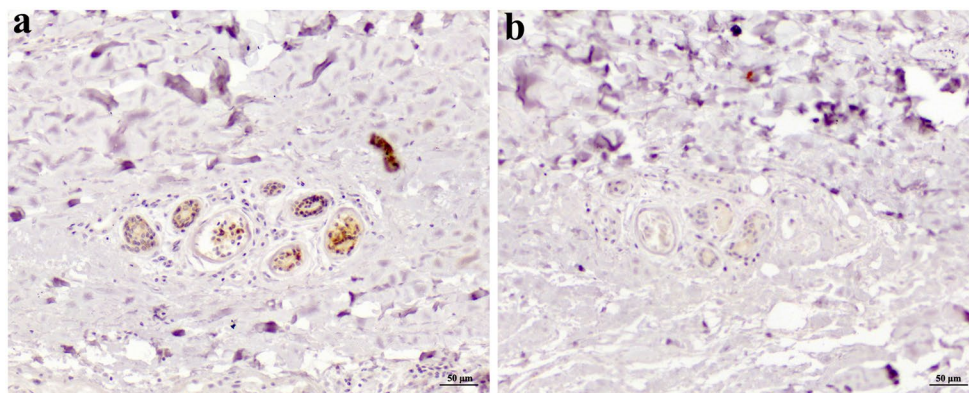


Fig. 4 Immunohistochemistry result of *Agkistrodon halys* venom **a** obvious positive expression in the subcutaneous sweat gland duct cells of the fang mark, indicating that *Agkistrodon halys* venom entered the subcutaneous sweat gland duct cells and accumulated there, immune-

stained with anti- *Agkistrodon halys* venom antibody (brown), bar = 50 μ m. **b** negative expression in the subcutaneous sweat gland duct cells of the fang mark, immune-stained without anti- *Agkistrodon halys* venom antibody, bar = 50 μ m

phenothiazines, and other common drugs, and all results were negative.

Forensic diagnosis

After a comprehensive analysis of the case information, pathological changes of multiple organs, and laboratory test results, it was believed that the deceased died of snake venom poisoning, bitten by an *Agkistrodon halys*.

Literature reviews

A systematic review of the literature was performed from 2000. Database search included PubMed and Web of Science. Keywords and combinations were related to “snake venom” or “*Agkistrodon halys*”.

Duplicate records were removed manually using Mendeley. Irrelevant titles or non-English articles were excluded. We then reviewed abstracts if available and full-text articles;

Table 1 Toxicity, components, and damage mechanisms of *Agkistrodon Halys* snake venom

Toxicity type		Related venom components	Mechanism
Hemotoxicity [2, 8]	Procoagulation	Thrombin - like enzymes	Activate coagulation factors (such as factor X and prothrombin), induce platelet aggregation, and the thrombin - like effect affects coagulation factors
	Hemolysis	PLA ₂ , SVSP [5]	Inhibit platelet aggregation, hydrolyze or bind to phospholipids, and degrade coagulation factors
	Bleeding	SVMP, SVSP [5, 9]	Affect capillary permeability
Neurotoxicity [2]	Presynaptic neurotoxicity	PLA ₂ [10]	Affect the presynaptic membrane and inhibit the release of neurotransmitters from the presynaptic membrane
	Postsynaptic neurotoxicity	PLA ₂	Affect the elimination of neurotransmitters and inhibit the binding of neurotransmitters to postsynaptic receptors
Histotoxicity [2]	Cytotoxicity	Cytotoxic PLA ₂ , β - defensin - like toxins [2]	Destroy the stability of the plasma membrane by affecting the lipid bilayer, or directly affect voltage - gated ion channels and membrane potential; destroy the lysosome membrane, leading to lysosome lysis and triggering apoptosis
	Extracellular matrix toxicity	SVMP, hyaluronidase [2]	Hydrolyze important components of the extracellular matrix, indirectly damage cells, and promote the diffusion of toxins

articles were excluded if they were irrelevant, not case reports, or did not identify snake envenoming as the cause of death. Studies involving editorials, technical notes, commentaries and opinion pieces were also excluded. After an initial assessment to determine the relevance of studies to the scope of the present review, 56 full-text articles were left.

Above all these articles, we reviewed, analyzed and summarized the toxicity, components, damage mechanisms,

and tissue pathological changes of *Agkistrodon halys* snake venom (Tables 1 and 2).

The snake venom of *Agkistrodon halys* is divided into hemotoxicity, neurotoxicity and histotoxicity. The hemotoxicity, which is the predominant poison of *Agkistrodon halys* venom, can effect the blood circulation through procoagulation, hemolysis and bleeding. The neurotoxicity can effect the nerve impulse transmission between synapses because

Table 2 Tissue pathological changes of *Agkistrodon Halys* bite

Injury site	Mechanism	Tissue pathological manifestations
Skin injury at the bite mark [2, 11]	Toxins induce the production of reactive oxygen species, stimulate tissues to release pro - inflammatory factors, and expand the inflammatory response	The epidermis and dermis of the bite - mark skin are separated, and the epidermis is defective. Acute cellulitis in the subcutaneous tissue of the bite mark: The collagen fibers in the dermis are swollen and disintegrated, and the interstitium has an infiltration of inflammatory cells mainly composed of neutrophils. The surrounding interstitium has edema with large - area bleeding and a large amount of fibrin exudation
Skeletal muscle injury [12–14]	Cytotoxicity directly or indirectly damages cells	Coagulation necrosis of muscle, disappearance of cross - striations, and enhanced eosinophilia
Acute kidney injury [12, 15, 16]	Cytotoxicity directly or indirectly damages cells; Hemotoxicity leads to DIC. Hemotoxic coagulation substances promote blood coagulation and thrombus formation, causing microcirculation disorders and inducing ischemic lesions; Activated PLA ₂ forms hemolytic toxins, causing intravascular hemolysis and resulting in hemoglobinuria; Rhabdomyolysis forms myoglobinuria. The above - mentioned factors lead to necrosis of the renal cortex or renal tubules, resulting in acute kidney injury	Toxic nephropathy: Renal tubular epithelial cells are necrotic and shed into the lumen, leaving the renal tubular basement membrane exposed. Some renal tubular epithelial cells proliferate, and red blood cell casts or protein casts are formed in some renal tubules
Acute liver injury	Cytotoxicity damages hepatocytes and liver parenchyma, causing acute liver injury; Hemotoxic coagulation substances promote blood coagulation and thrombus formation, causing microcirculation disorders and inducing ischemic lesions, resulting in hepatocyte damage	Toxic hepatitis: The structure of the liver lobules is disordered, the hepatic cell cords are broken, and the hepatocytes have coagulation necrosis
Myocardial damage [17]	Cytotoxicity damages cells; Hemotoxicity leads to circulatory disorders, causing myocardial damage	The sarcoplasmic cross - striations of necrotic myocardial cells disappear, and eosinophilia is enhanced
Blood system [2, 12, 18]	Hemotoxicity affects blood coagulation through pro - coagulant or anticoagulant methods, leading to coagulation disorders and even DIC	Microcirculation disorders in multiple organs, intravascular thrombus formation, and ischemic lesions of organs
Nervous system [2]	Neurotoxins affect synapses, leading to muscle transmission block	Respiratory muscle paralysis, hypoxia, and carbon dioxide retention

of which we can divide the neurotoxicity into presynaptic neurotoxicity and postsynaptic neurotoxicity. Some components of the snake venom can damage tissue through the toxicity to cell or extracellular matrix, which is called histotoxicity. All these types of toxicity can cause skin injury, skeletal muscle injury and organ injury, such as acute kidney injury and acute liver injury.

Discussion

Agkistrodon halys, a member of the Viperidae family, produces mixed venom with predominant hemotoxicity. After being bitten by *Agkistrodon halys*, the snake venom is injected into the body through the venomous teeth and quickly spreads throughout the body through the lymphatic and circulatory systems, damaging blood vessels, muscles, and red blood cells, causing coagulopathy, promoting tissue damage, edema, necrosis, hemostatic imbalance, bleeding, and acute renal failure [19, 20]. The main mechanism of death from *Agkistrodon halys* snake venom poisoning is respiratory and circulatory failure, and acute renal failure may also lead to death.

In addition to the tissue and organ pathological changes caused by snake bites, the direct qualitative and quantitative detection of snake venom is the key to the forensic identification of snakebite cases. There are relatively mature detection techniques applied in forensic reports of related snake bites. For example, Huang et al. [21] utilized a sandwich ELISA method to qualitatively and quantitatively detect snake venom in a suspected snake - bite case, assisted in clarifying that the type of snake bite in the case was *Deinagkistrodon acutus*, and quantitatively analyzed the content of snake venom. This approach aided in determining that the type of snake and facilitated the quantitative analysis of venom content, which elucidate the post-mortem distribution and concentration of the venom. Tian et al. [7] refined this detection method and applied it to the detection of suspected *Bungarus multicinctus* bite cases, directly and accurately clarifying the type and content of snake venom. This provided robust evidence for forensic identification.

In this case, suspected venomous snake fang marks were found on the inner side of the right foot arch of the deceased. There was edema and hemorrhage in the right foot and lower limb. Histopathological findings revealed hemorrhagic and necrosis at the corresponding site, with toxic pathological alterations observed in organs such as the liver and kidneys. Referring to relevant literature, we employed a combination of sandwich ELISA and immunohistochemistry to detect the type and the content of snake venom, leading to the comprehensive conclusion that the cause of death was attributed to a bite from *Agkistrodon halys*.

The results of the sandwich ELISA indicated that, among the assays conducted for various snake venoms, only the *Agkistrodon halys* plate yielded a strong positive result in the subcutaneous tissue at the suspected fang mark on the right foot arch. The test results were relatively stable in multiple tissue organs, and the OD value was high. Considering comprehensively, ELISA is a good detection method for suspected *Agkistrodon halys* snake venom cases, and the test results are reliable.

The IHC analysis revealed an obvious positive expression in the subcutaneous sweat gland duct cells of the fang mark, indicating that after being bitten by *Agkistrodon halys*, in addition to entering the body through the circulatory system, the snake venom was also likely to enter the skin sweat glands and accumulate there. In some literature using IHC to detect snake venom, positive snake venom particles are mostly found in the inflammatory cells of the fang-mark skin or in the main snake-venom-accumulating organs such as the liver and kidneys, and positive expression is rarely seen in the skin sweat glands.

Agkistrodon halys is one of the most prevalent venomous snakes in China. Its bites are relatively common in snake-bite-related fatal cases in China. Most cases are accidents, but there are also some cases of suicide or homicide by injecting snake venom [22, 23]. In the forensic identification practice of snake-venom-poisoning cases, it mainly relies on the combination of the case history, presence of fang marks, histopathology, and snake-venom detection [21]. We hope that the autopsy results, immunohistochemistry, ELISA results reported here, and the literature review can provide help for forensic practice and further research, and provide certain diagnostic basis for future forensic identification of suspected snake-bite cases.

Conclusion

In this case, a combination of ELISA and immunohistochemistry methods was employed to directly conduct qualitative and quantitative analysis of the snake venom, which helped confirming the snake as *Agkistrodon halys*. The concentration of the snake venom was determined to be 6.58 ng/mL, and it was ascertained that the venom had entered into the skin sweat glands, accumulating in these specific regions through the aforementioned method. Furthermore, a review of the literature was conducted to summarize the forensic characteristics associated with this type of snake venom poisoning, aiming to provide reference for the forensic identification.

Author contributions Yubei Wang: Literature search, analysis and manuscript writing; Zhaoxuan Wang: Literature search and sorting; Xiangting Gao: Literature review and data analysis; Qing Shi: Literature

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Declarations

Ethical approval The informed consent of the deceased's family has been obtained and relevant legal documents have been signed. The autopsy was conducted in strict accordance with Forensic medicine — Specifications for examination of poisoning deaths (China, GA/T 167–2019) to ensure compliance with ethical requirements. All data and sample collections were approved by the Ethics and Morality Committee of Huazhong University of Science and Technology (ethical approval number: 2024BL151).

Conflict of interest The authors declare they have no conflict of interests.

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